08: Linkers + Logic Analyzers

Engr 315: Hardware / Software Codesign

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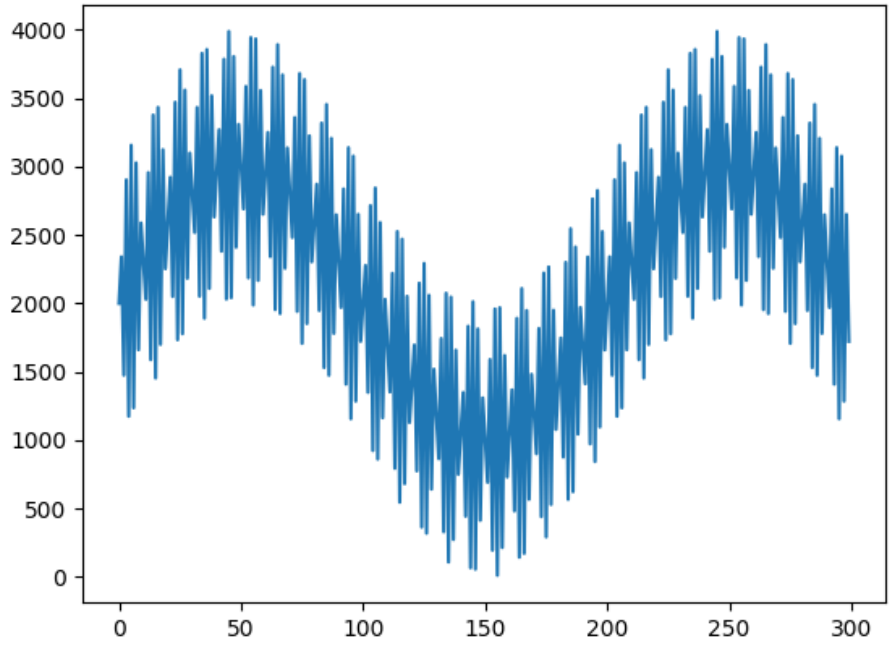
Let’s talk P3

- New Version: (V7?) posted soon
- ILA tutorial finalized
- Bug with SDK code fixed.
- Absolute path probems in SDK “/home/lukefahr’.

P3: Filtering with DMA



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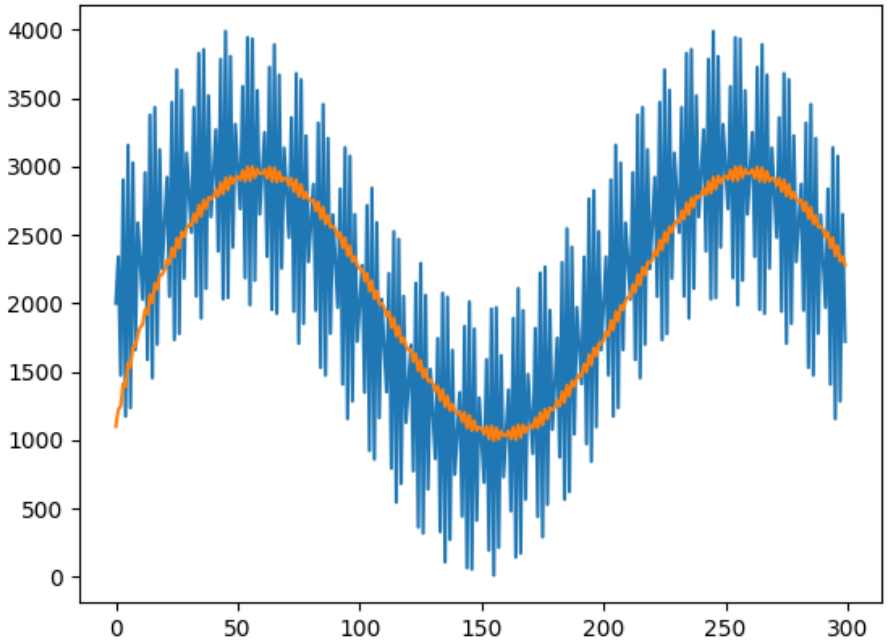


Magic Incantations makes things work!

```
uint32_t PING [MAX_PKT_LEN] __attribute__ ((section(".dma_data")));
uint32_t PONG [MAX_PKT_LEN] __attribute__ ((section(".dma_data")));

...

XAxiDma_TxRxBuffer( … ,  PING, PONG, sizeof(PING));
```

Familiar Compilation Steps:

- `vi main.c`
- `gcc main.c –o program.out`
- `./program.out`

What’s actually going on:

Let’s see an example

main.c

```
#include <stdio.h>
#include <stdlib.h>
#include "foo.h"
int main()
{
    printf("Hello World\n");
    int a = foo(2,3);
    printf("A: %d\n", a);
    return 0;
}
```

foo.c

```
#include <stdio.h>
#include <stdlib.h>

int foo( int x, int y)
{
    int z = x + y;
    return z;
}
```

Compilation

```
gcc -g -O0 -S main.c

-g:  debugging symbols
-O0:  Optimization level 0
-S:  emit assembly
```

Assembly

```
as main.s -o main.o
```

```
-o:  output file name
```

Linking

```
ld main.o foo.o
   -o program.elf*
```

-o: output binary name

* Requires extra arguments for OSX

Execution in Linux

```
./program.elf

> Hello World
> A: 5
```

ELF: Executable Linkable Format

Let’s look at an ELF:

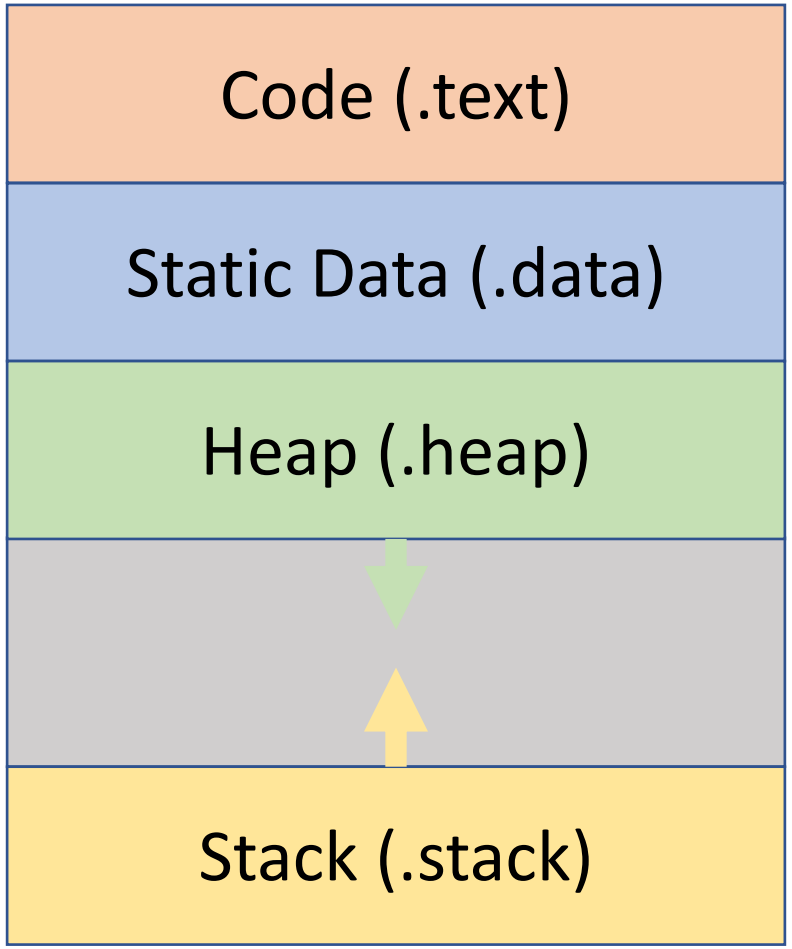
```
objdump –DS program.elf >
                        program.dis
```

What if there’s no Linux?

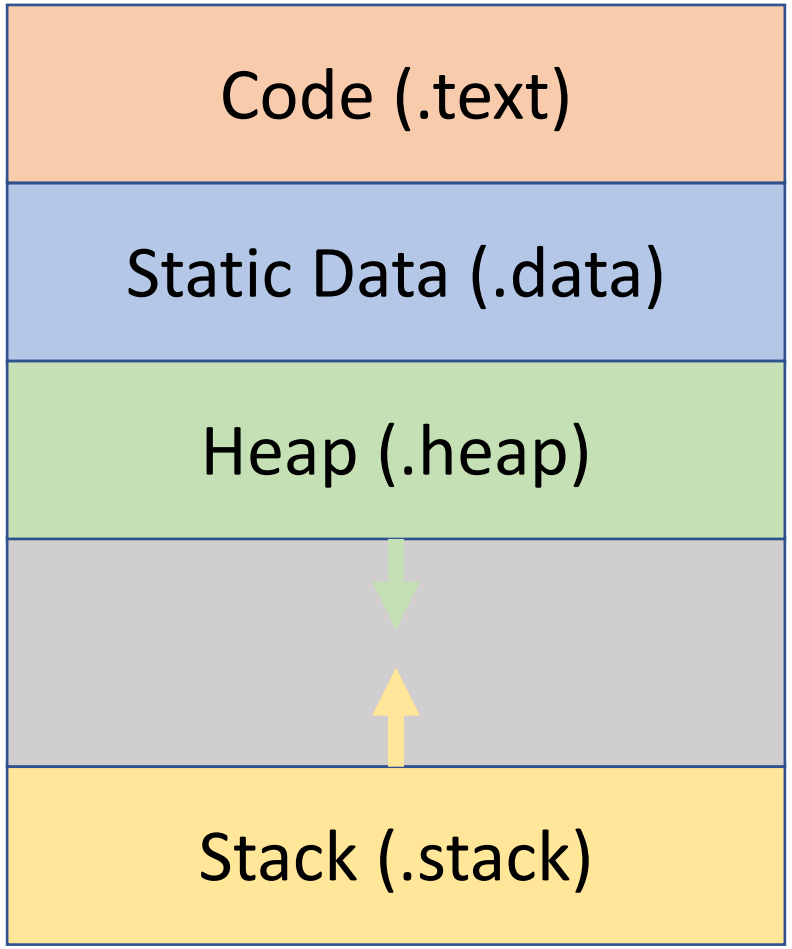
- Elfs are not enough.
- We need to build a full binary file:

```
objcopy -O binary program.elf
                              program.bin
```

Memory Layout



How does the compiler/linker know where to put stuff?



Linker Scripts

Filename: `src/ldscript.ld`

```
/* memory layout for an STM32F401RE */

MEMORY
{
    FLASH (rx)  : ORIGIN = 0x08000000, LENGTH = 512K
    SRAM (xrw)  : ORIGIN = 0x20000000, LENGTH = 96K
}

ENTRY(main)

/* output sections */
SECTIONS
{
    /* program code into FLASH */
    .text : ALIGN(4)
    {
        *(.vector_table)       /* Vector table */
        *(.text)               /* Program code */
        KEEP(*(.vector_table))
    } >FLASH

    /* Initialized global and static variables (which
       we don't have any in this example) into SRAM */
    .data :
    {
        *(.data)
    } >SRAM
}
```

Let’s check out P3

- Binary ELF
- Linker Script
- And where .dma_data is in it.

Logic Analyzers

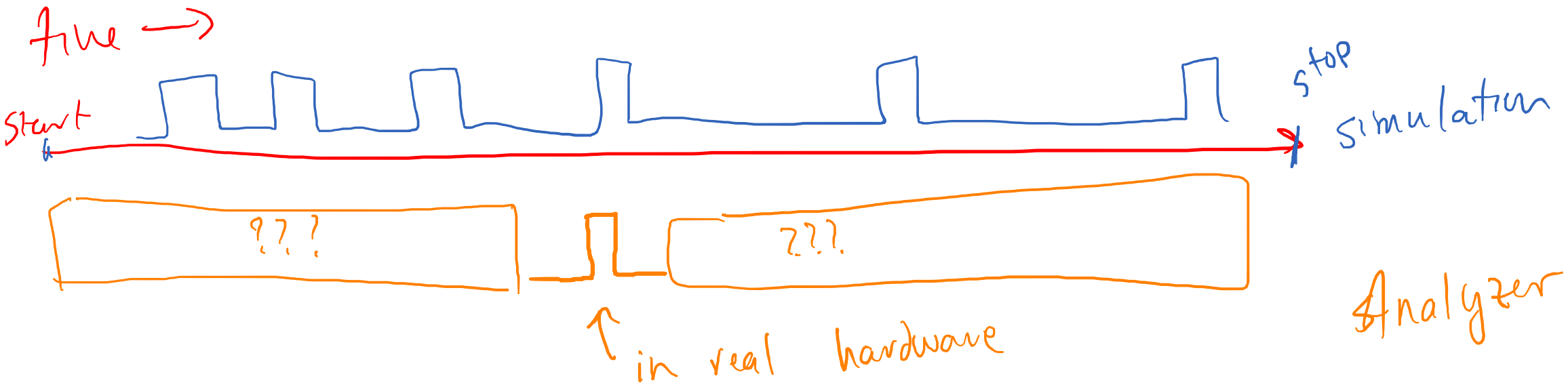
Logic Analyzers

- A **logic analyzer** is an electronic instrument that captures and displays multiple signals from a digital system [wiki]

- Captures and displays signals from ***inside*** the FPGA.

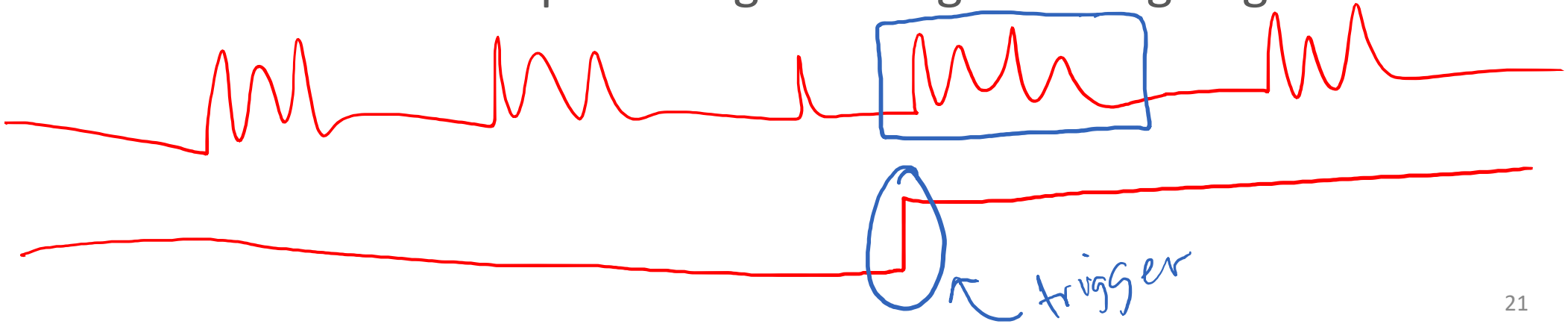
Simulation vs. Analyzers

- Simulation: what *should* happen inside the FPGA for all time simulated
- Analyzer: what *did* happen inside the FPGA for a small time window



Logic Analyzers let you:

- Capture a small window of real-live signals
- Window centered around a specific event, called a **trigger**
- Often specific signal rising or falling edge



Logic “Trigger”

Xilinx “In Logic Analyzers”

- Uses part of the FPGA to monitor another part.

Let’s look at P3’s Logic Analyzer

- Setup Trigger

Next Time:

- Performance Analysis

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